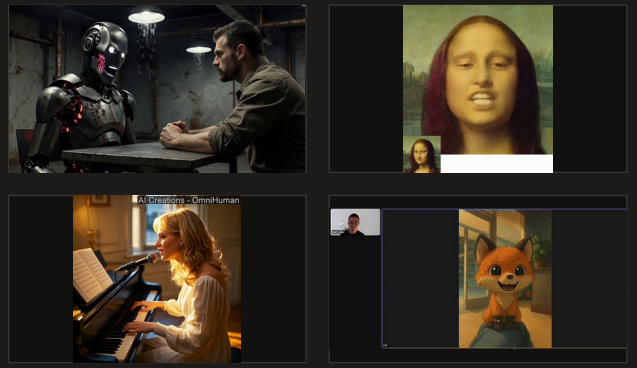
AMCC 5160 · LECTURE 08 · 20 OCTOBER 2026



When AI Video *Learned to Talk*

Sound, performance, and real-time characters · Fall 2026

Harry Yang · Division of Arts and Machine Creativity, HKUST · MIL13002

Sound off: what are they saying?



Play it muted first. Guess the line, the voices and the room · Ovi · Character.AI · click to watch · source ↗

Sound on: one model made all of it



Voices, room tone and timing were generated together with the picture · Ovi · Character.AI · click to watch · source ↗

Four stories from this year

01

Sound is no longer post-production

Picture, voice, effects and music now come out of one model

Veo 3 (May 2025) → Kling 4.0 (Oct 2026)

02

Directing a performance, not a prompt

Lip-sync grew into acting: emotion, timing, gesture, duets

Wav2Lip (2020) → OmniHuman-1.5 (2025)

03

Characters that listen

Real-time avatars that react while you are still talking

Avatar Forcing, CVPR 2026

04

Who is the performer?

Synthetic actors, digital replicas, and the new rules

Tilly Norwood → SAG-AFTRA 2026

Where we've been: seven weeks in one slide

01 · Sep 1

Neural networks to AI filmmaking

02 · Sep 8

Generative editing and art direction

03 · Sep 15

Inside the generator

04 · Sep 22

AI video and art filmmaking

05 · Sep 29

Adaptation and authorship

06 · Oct 6

Quantization for faster inference (Yaofu)

07 · Oct 13

Music, minutes, and machines

08 · Tonight

Sound, performance, real-time characters

Tonight reuses three tools: tokens (Week 3), speed tricks (Weeks 6–7), and authorship (Week 5).

Synchresis: why sound is half the picture

The idea

Film theorist Michel Chion (Audio-Vision, 1990): when a sound and an image happen at the same moment, the brain welds them into one event, even if they were recorded apart.

Why Foley works

Celery snapped for a breaking bone, coconut shells for horse hooves: the audience believes it, because the timing is right.

Why AI lip-sync fails loudly

The same welding makes a mismatch jump out. Broadcast tests (ITU-R BT.1359): viewers detect sound about 45 ms early or 125 ms late.

Keep this word. It explains both the magic and the failures you will see tonight.

STORY 1 OF 4

01

Sound is no longer post-production

Picture, voice, effects and music from one model, in one pass



Veo 3 · Google DeepMind

The old pipeline: five tools, five chances to drift

Picture

Text-to-video makes a silent clip

→

Voice

Text-to-speech or a recorded actor reads the line

→

Lips

A lip-sync model (e.g. Wav2Lip, 2020) repaints the mouth

→

Foley

Video-to-audio (e.g. MMAudio) adds footsteps and room

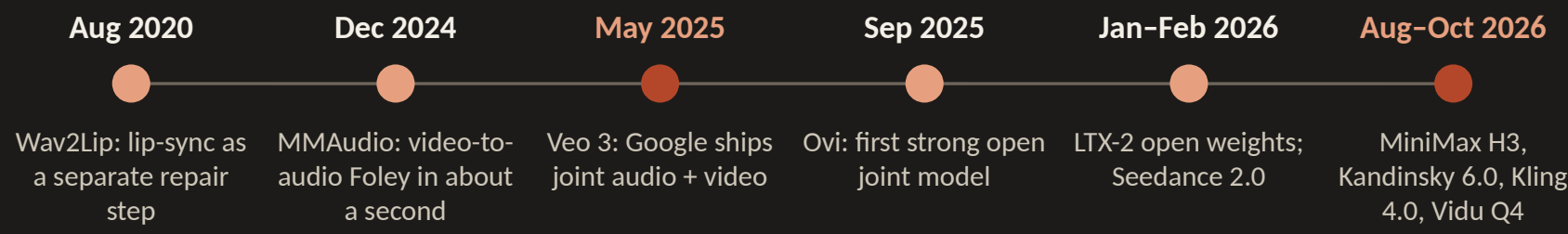
→

Mix

An editor balances music, dialogue and effects

Each hand-off can lose timing, emotion or identity. This is still how many of you made Assignment 1.

Eighteen months, from add-on to native



By mid-2026, almost every frontier video model generates its own sound.

Veo 3: the clip that started it



Voice, wind and sea from one prompt, in one pass · Veo 3 · Google DeepMind · click to watch · source ↗

Source: Google DeepMind, Veo 3 model card (2025)

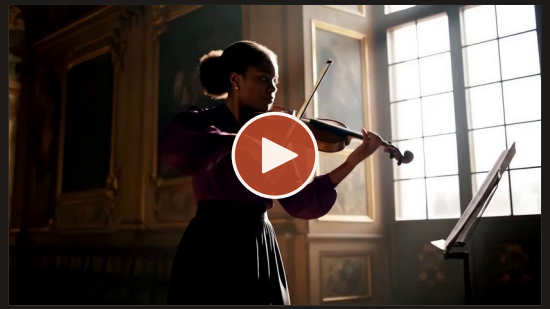
Three kinds of sound, one model



Effects

Sizzle and clatter timed to each toss of the wok.

Veo 3 · Google DeepMind · click to watch · source ↗



Music

The bow and the note move together.

Veo 3 · Google DeepMind · click to watch · source ↗



Dialogue

A detective interrogates a rubber duck. Comedy needs timing.

Veo 3 · Google DeepMind · click to watch · source ↗

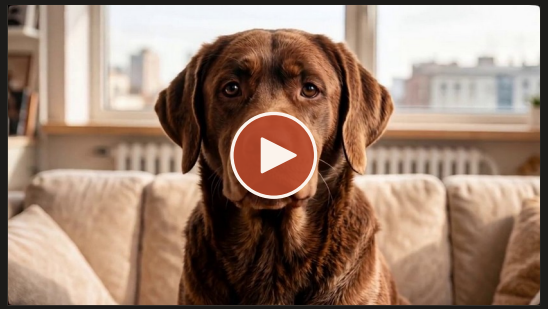
Listen for synchresis: which sound "welds" to the picture best?

Who ships native audio in 2026

Model	Maker	Open?	What stands out
Veo 3.1	Google	Closed	Audio in reference, first/last-frame and Extend modes
Seedance 2.0 / 2.5	ByteDance	Closed	Up to 9 images + 3 videos + 3 audio clips as references
Kling 4.0	Kuaishou	Closed	Up to 30 s per generation (Sep 28 preview)
Vidu Q4 Preview	ShengShu	Closed	Voice references for consistent characters (Oct 7)
LTX-2 / 2.5	Lightricks	Open weights	19B: 14B video + 5B audio stream
Ovi	Character.AI	Open (Apache-2.0)	Twin towers; audio tower trained from scratch
MiniMax H3	MiniMax	Open weights	33B single stream; 32 kHz stereo audio
Kandinsky 6.0	Kandinsky Lab	Open (MIT)	3B Lite and 29B Pro; 44 kHz audio

Sources: official model cards, technical reports and release notes (Google, ByteDance Seed, Lightricks arXiv 2601.03233, Character.AI arXiv 2510.01284, MiniMax, Kandinsky arXiv 2610.05608).

Open weights, this summer and autumn



Kandinsky 6.0 (Oct 2026)

MIT license; 3B Lite and 29B Pro; 44 kHz audio.

Kandinsky 6.0 · Kandinsky Lab · click to watch · source ↗



MiniMax H3 (Aug 2026)

33B single stream; 32 kHz stereo; 11 languages of dialogue.

MiniMax H3 · MiniMax · click to watch · source ↗



NAVA (May 2026)

6.3B, Apache-2.0; best open lip-sync on Verse-Bench.

NAVA · Baidu ERNIE · click to watch · source ↗

You can download and run all three; check each license before commercial use.

Closed models: Seedance 2.0's range



A painting comes alive

Chinese ink-painting style, with hooves and wind generated to match.

Seedance 2.0 · ByteDance Seed · click to watch · source ↗



A concert hall

Hands, music and the room's reverb, together.

Seedance 2.0 · ByteDance Seed · click to watch · source ↗

Under the hood: what is sound to a model?

48,000

Numbers per second per channel: raw audio is a pressure wave, sampled

→

Mel

Many short FFTs turn it into a picture of pitch over time (a spectrogram), warped to how we hear

→

25 / s

An audio VAE squeezes it to about 25 latent vectors per second (LTX-2: 128 numbers each)

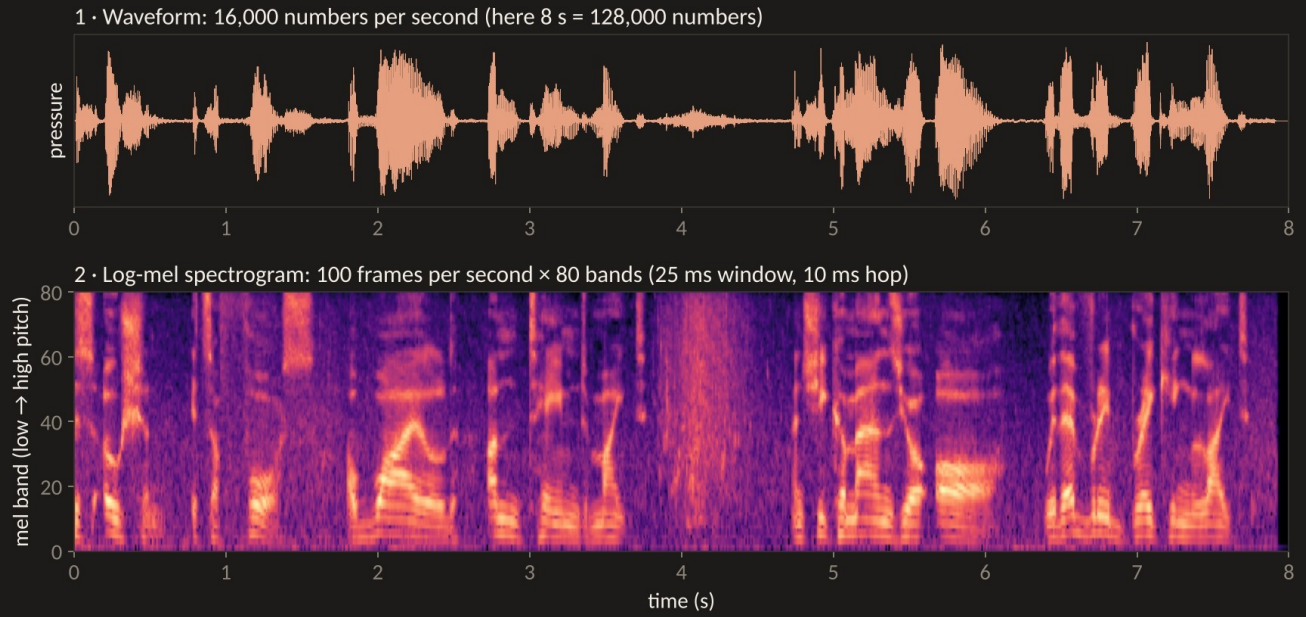
→

Tokens

Now sound is a short sequence the transformer can attend to, like video tokens

Adapted from CMU 10-423 Generative AI, Lecture 24 “Audio Understanding and Synthesis” (Gormley & Nayebi, Fall 2025); numbers from LTX-2 (arXiv 2601.03233).

See it: 8 seconds of the Veo 3 sailor, as a model sees it



Computed from the demo clip for this lecture. Horizontal stripes are the harmonics of a voiced vowel; vertical bursts are consonants and splashes; the hiss between words is wind and sea. A mel spectrogram is literally an image of sound.

Why the two streams are not the same size

~21,600

Video tokens per second (720p, 24 fps, Wan-style VAE: our Week 7 worked example)

25

Audio latents per second in LTX-2 (128 numbers each)

14B + 5B

LTX-2 spends most of its parameters on picture, a smaller tower on sound

Picture is dense in space; sound is dense in time. Hence “asymmetric dual-stream” designs.

Four ways to wire sound and picture

Picture → sound

Generate the video, then a second model listens to it and adds sound. MMAudio (CVPR 2025), HunyuanVideo-Foley. Good Foley, but the picture never hears the sound.

Sound → picture

Start from audio and animate a face or body to it. Talking-head models: VASA-1, OmniHuman, InfiniteTalk. Story 2.

Two towers that talk

A video network and an audio network denoise together and exchange information in every block. Ovi, LTX-2, Kandinsky 6.0.

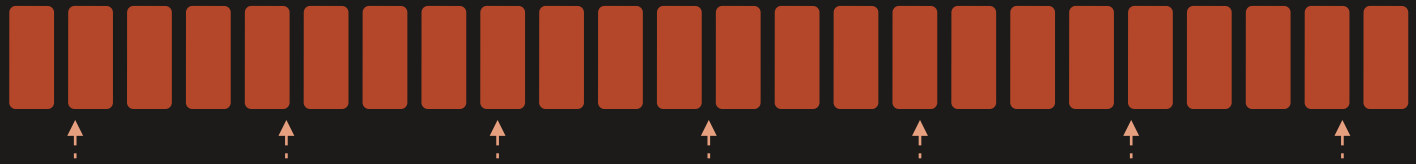
One stream

Put text, video and audio tokens in one sequence; one transformer attends to all. MiniMax H3 (33B), daVinci-MagiHuman (15B).

Inside LTX-2: two streams, one clock

Video stream

14B · 3D RoPE (x, y, t)
about 20,000 tokens per second of 720p



Bidirectional cross-attention in every block, using only the time positions

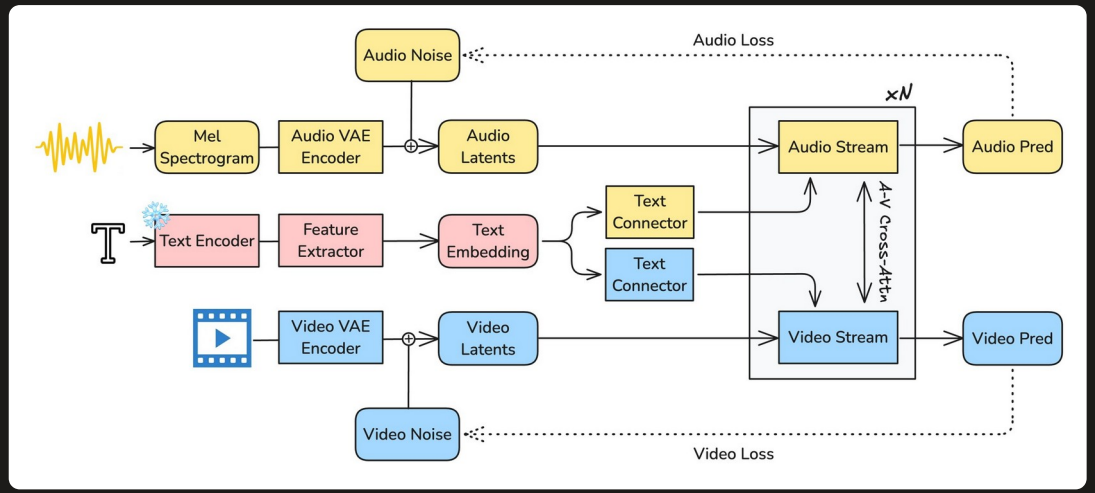


Audio stream

5B · 1D RoPE (t)
about 25 tokens per second

Both streams also share the diffusion timestep. Result: up to 20 s of video with stereo sound; 1.22 s per step on an H100 vs 22.3 s for Wan 2.2-14B. Source: LTX-2, arXiv 2601.03233.

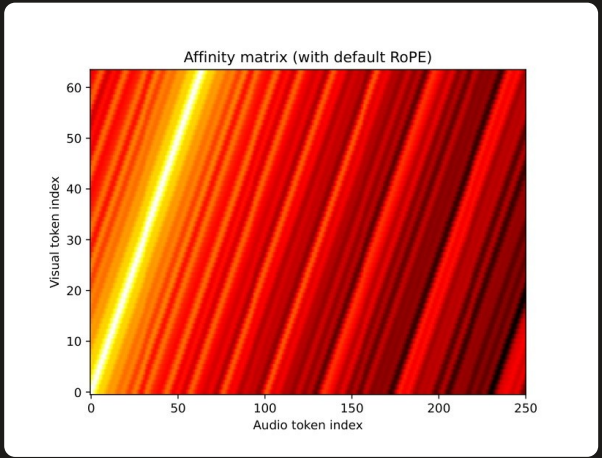
From the technical report: LTX-2 at a glance

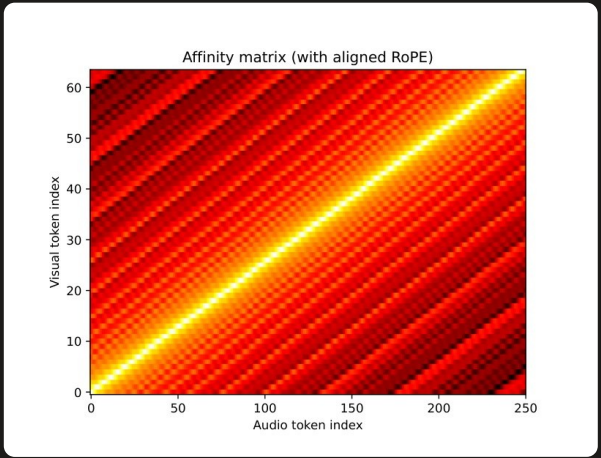


- Two VAEs: one compresses video, one compresses the mel spectrogram
- A text encoder (Gemma 3) feeds both streams through separate “connectors”
- The streams denoise together; the A ↔ V cross-attention is the bridge
- Two losses at the end: one for picture, one for sound

Figure: LTX-2 technical report, Lightricks, arXiv 2601.03233 (2026). Shown for teaching.

Keeping time: why positions must be aligned





Default positions

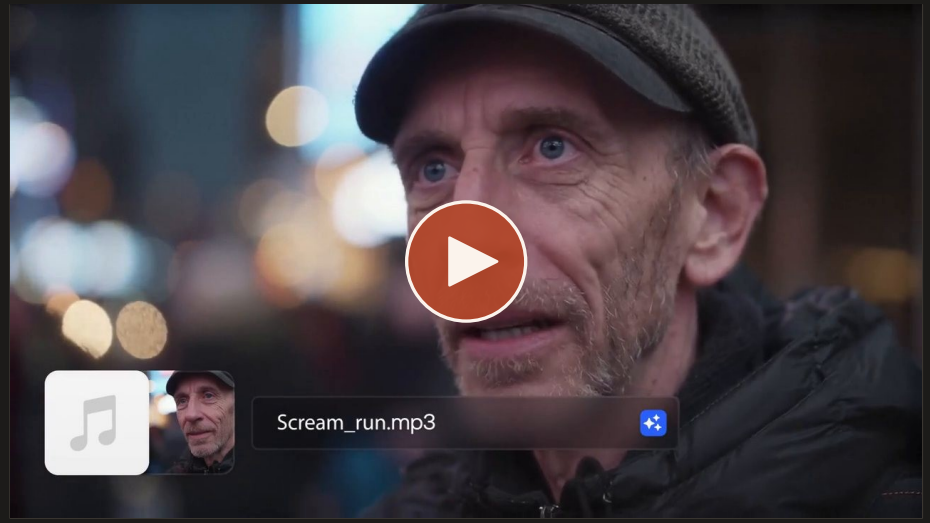
Video and audio tokens arrive at different rates, so “nearest in position” drifts away from “same moment”.

Aligned positions

Scale the positions to seconds and the bright diagonal lines up: each frame attends to its own moment of sound.

Figures: MMAudio (Cheng et al., CVPR 2025), arXiv 2412.15322. LTX-2 and Ovi use the same idea: only time positions are shared across streams.

LTX-2: give it a sound, get a performance



LTX-2 · Lightricks · click to watch

- Audio-to-video: the scream comes first, the face is generated to match it
- Open weights (LTX-2 Community License); runs on one large GPU
- Same model also does text-to-audio+video and image-to-audio+video
- Under the hood: the two-stream design from the previous slide

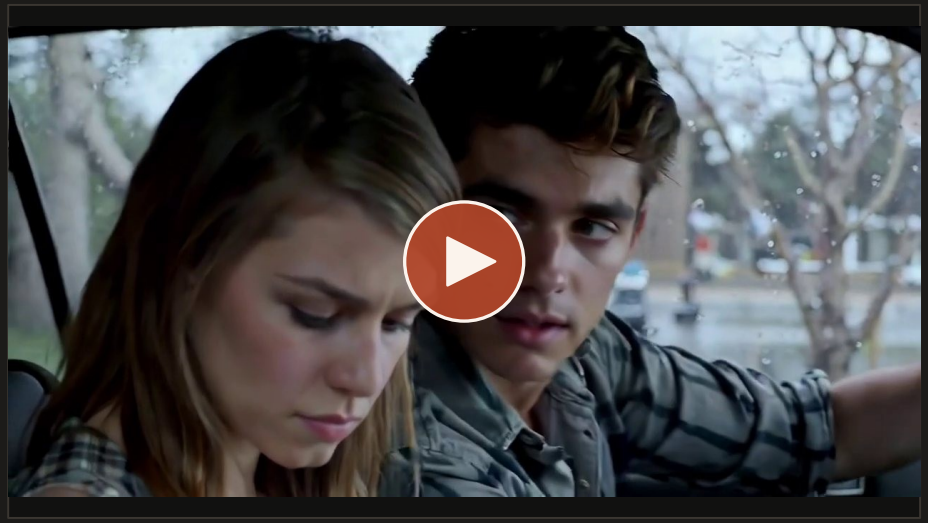
▶ LTX-2 on GitHub

github.com/Lightricks/LTX-2

▶ Technical report (arXiv)

arxiv.org/abs/2601.03233

Ovi: two voices, one take



Ovi · Character.AI · click to watch

- Twin towers of the same shape: video from Wan 2.2, audio trained from scratch
- Dialogue written in the prompt; the model decides who speaks when
- 5 or 10 s, 24 fps, 960×960; Apache-2.0
- Released by Character.AI and Yale, Sep 2025

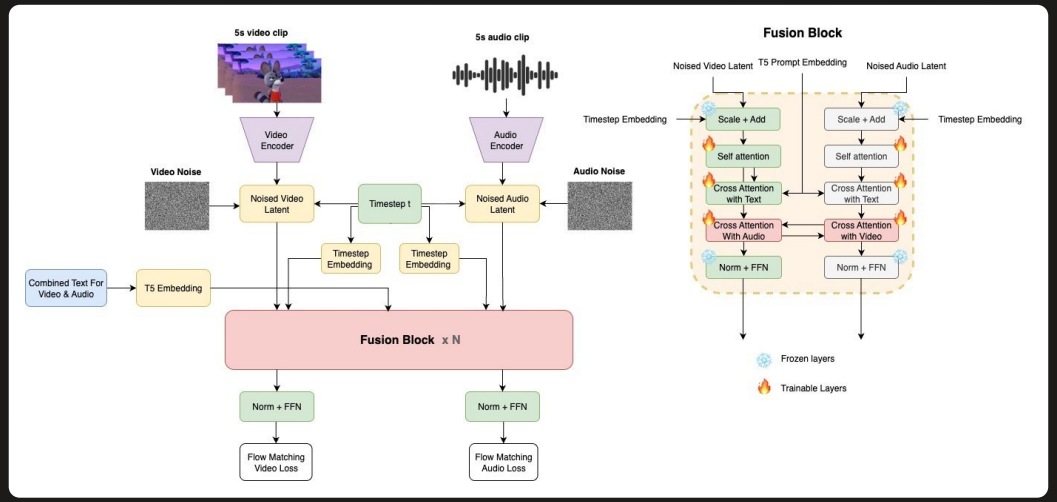
▶ **Ovi project page**

aaxwaz.github.io/Ovi

▶ **Paper (arXiv 2510.01284)**

arxiv.org/abs/2510.01284

Ovi’s twin towers



- Left tower: video (initialised from Wan 2.2). Right tower: audio (trained from scratch)
- Same shape, block for block, so they can talk at every layer
- Fusion block: self-attention, then cross-attention to text and to the other tower
- Frozen pieces (snowflakes) vs trained pieces (flames): reuse what already works

Figure: Ovi, Character.AI and Yale, arXiv 2510.01284 (2025). Shown for teaching.

Recipe: how to teach a video model to hear

1 · Reuse

Start from a strong video model (Kandinsky 5.0, Wan 2.2)

→

2 · Listen

Train the audio tower alone on text-to-audio first: ~40M tracks for Kandinsky 6.0

→

3 · Join

Train both together on clips with sound: ~7M filtered segments

→

4 · Polish

Fine-tune, reinforcement learning, then distil to fewer steps (10 for Kandinsky 6.0)

Source: Kandinsky 6.0 Video, arXiv 2610.05608 (Oct 2026). NAVA (Baidu, 6.3B) follows a similar path from Wan 2.2: audio-heavy first, then joint.

Data: the hardest part is finding clips that are in sync

Throw most of it away

Web video often has music pasted over it, dubbed voices, or a voice-over. Kandinsky keeps a person-on-screen clip only if a sync score says the lips match.

Caption the sound too

Captions describe voices, effects and music, and put spoken words in quotes, so prompts can do the same.

Know who is talking

NAVA ties a speaker's voice embedding to each line of speech, so two characters keep two voices.

Kandinsky 6.0: ~7M segments kept, about a quarter with no faces at all. NAVA: ~15M clips, ~7 s on average, 160 H100s for ~4 weeks.

Anatomy of an audio-video prompt

Layer	What to write	Example
Picture	Shot, subject, light, camera	Close-up, rainy night, a taxi driver turns to the back seat
Dialogue	Who speaks, the exact words in quotes, how	He says quietly, tired: “You sure this is the address?”
Effects	Sounds tied to visible actions	Wipers squeak once; a turn signal ticks
Ambience	The room or place, always on	Rain on the roof, distant traffic
Music	Inside the scene or score? Mood, level	Low radio jazz from the dashboard, under the voice

Tip from Seedance users: keep a spoken line under about 15 s, and avoid asking for big head turns while a character speaks.

Copy these: three prompts that work

Dialogue scene (Veo, Seedance, Kling, LTX-2, Ovi)

```
Medium close-up, night, inside a taxi in the rain. The driver, 50s, tired, turns to the back seat
and says quietly: "You sure this is the address?" Wipers squeak once. Rain on the roof. No music.
```

Sound-led shot (effects and ambience, no speech)

```
Slow push-in on a wok over a high flame. Oil hisses; each toss of the vegetables lands with a
metallic clatter, in time with the motion. Busy restaurant kitchen ambience in the background.
```

Music inside the scene

```
A street saxophonist plays a slow jazz phrase on a wet city corner at dusk; footsteps and traffic
pass behind him. The music is played on screen; camera holds on his fingers for the last note.
```

Pattern: picture → who says what (in quotes) → effects tied to actions → ambience → music, and say “No music” when you mean it.

Be the Foley artist: what should this sound like?



First muted: list every sound you would add. Then compare with the model · Ovi · Character.AI · click to watch · source ↗

Listen twice: which Foley is better?



Version A

Same burning grassland, sound added by one video-to-audio model.

Clip A · click to watch · source ↗

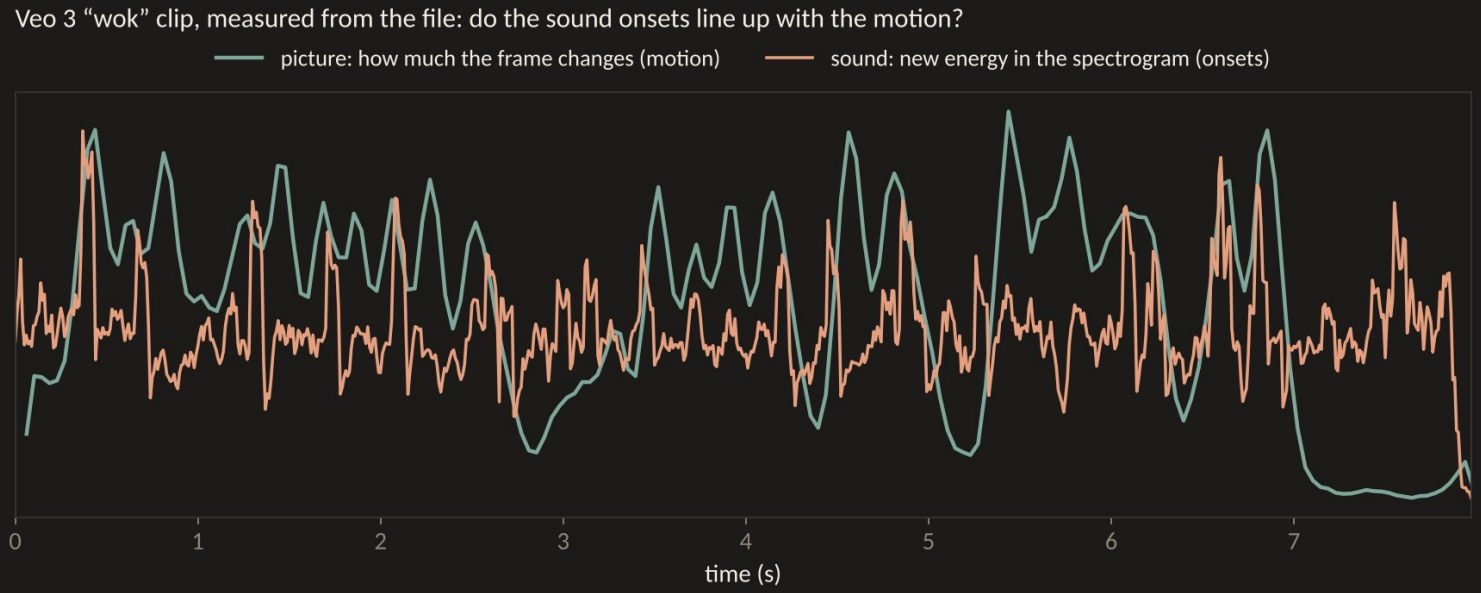


Version B

Same picture, sound from a different model.

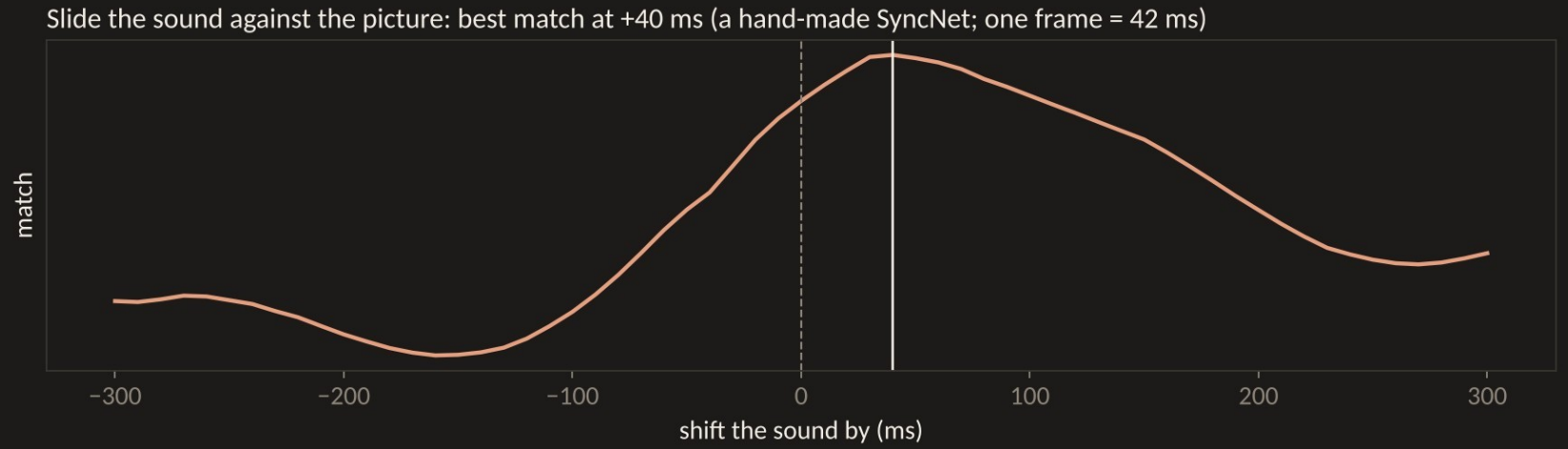
Clip B · click to watch · source ↗

Measure it: do the sounds line up with the motion?



Computed from the Veo 3 “wok” clip. Green: how much each video frame changes. Orange: new energy in the sound’s spectrogram (onsets). Many peaks coincide, which is what synchresis needs.

Slide the sound: a hand-made SyncNet



Shift the sound in 10 ms steps and score how well the two curves match. The best match is about one frame (42 ms) from zero, inside the range viewers tolerate (45 ms early to 125 ms late).

Diegetic or not? Music that lives in the scene



Inside the scene (diegetic)

A street saxophonist: the characters could hear this music too.

Ovi · Character.AI · click to watch · source ↗



Played on screen, generated together

UniVerse-1 stitches a video expert and a music expert into one model.

UniVerse-1 · click to watch · source ↗

Where it still breaks

Two people talking

The wrong face moves, or both mouths move. ByteDance lists multi-person lip-sync as an open problem.

Motion vs. mouth

Ask for a head turn or a laugh mid-sentence and the lips lose the words.

Audio artefacts

Metallic voices, clipped words, music that swells over dialogue.

Accents and languages

Lip-sync quality varies a lot by language; dialects are harder.

Exact events

A glass breaks, but the crash comes a few frames early.

Same voice, next shot

A character's voice changes between generations unless you give a voice reference.

Can you run it yourself? Speed, again

18×

LTX-2 vs Wan 2.2-14B per diffusion step on one H100 (1.22 s vs 22.3 s)

35×

OmniForcing: a 5-s clip in 5.7 s instead of 197 s, by distilling LTX-2 into a streaming model

~25 fps

OmniForcing streams audio + video on one GPU, first chunk in ~0.7 s

Same tricks as Weeks 6–7: fewer steps (distillation), chunks + KV cache, fewer bits. Sources: LTX-2 arXiv 2601.03233; OmniForcing arXiv 2603.11647.

For sound designers and composers: what changes?

What the model now does

- A first-draft mix, already in sync
- Room tone, ambience, footsteps “for free”
- Diegetic music that matches the instrument on screen
- Temp dialogue for animatics and pitches

What stays human

- Choosing what NOT to hear: silence, focus, point of view
- The exact event: the hit, the cut, the joke’s timing
- Score: emotion that the characters cannot hear
- Taste, and consent for whose voice is used

FOR THE ROOM

If the model writes the sound with the picture, who is the sound designer now?

And would you credit the model, the prompt writer, or the people whose recordings trained it?

STORY 2 OF 4

02

Directing a performance, not a prompt

From “do the lips match?” to “is this good acting?”



What is a performance made of?

Voice

Pitch, pace, breath, accent, the cracks in it

Timing

Pauses, interruptions, the beat before the joke

Mouth

Lips, jaw and teeth matching every sound

Face and eyes

Brows, blinks, where the gaze goes, micro-expressions

Body

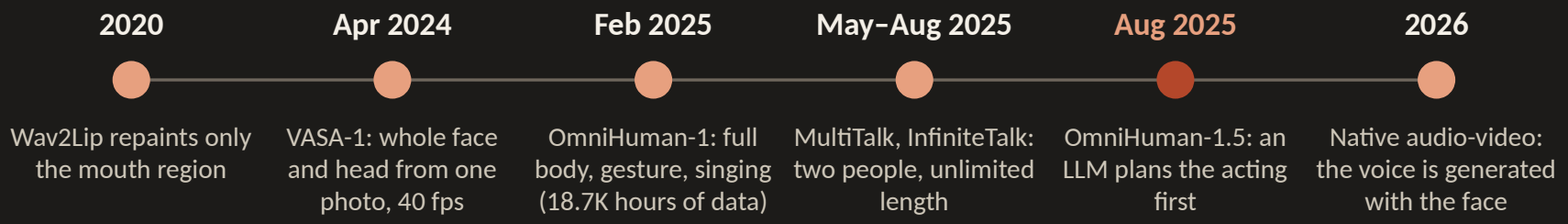
Head motion, shoulders, hands, posture

Listening

What you do while the other person talks

Early models did only the mouth. 2026 models attempt all six; listening is the newest (Story 3).

From lip-sync to acting



Sounds you can see: phonemes → visemes

Sound group	Examples	What the mouth does	Easy to fake?
Bilabials	/p/ /b/ /m/ — “map”, “Bob”	Lips fully close, then release	No: viewers check these
Labiodentals	/f/ /v/ — “five”	Upper teeth touch lower lip	Hard
Rounded vowels	/u/ /o/ — “who”, “go”	Lips push forward into an O	Medium
Open vowels	/a/ — “father”	Jaw drops wide	Easy
Tongue sounds	/t/ /d/ /k/ /g/	Mostly hidden inside the mouth	Easy: invisible

Many sounds share one mouth shape (a viseme). A model must hit the visible ones exactly, especially lip closures.

Under the hood: how a voice drives a face

Voice

16 kHz audio: the line to be spoken

→

Audio encoder

A pretrained speech model (wav2vec 2.0 or Whisper) gives one feature vector per video frame

→

Cross-attention

Each frame's tokens look up "what sound is happening now?" inside the video DiT

→

Frames

Mouth, face and body follow the sound; a reference photo keeps identity

Used by VASA-1 and OmniHuman (wav2vec), LatentSync and MuseTalk (Whisper), Avatar Forcing (wav2vec 2.0). The speech model already "knows" phonemes, so the video model does not need to learn them from scratch.

Under the hood: how we measure lip-sync (SyncNet)

5 frames

Crop the mouth: 0.2 s of video (25 fps), plus the matching 0.2 s of audio

→

Embed

SyncNet (Oxford, 2016) maps both to vectors; in sync = close together

→

Slide

Shift the audio ±15 frames and find where the distance is smallest

→

Score

Sync-D = that smallest distance (lower is better). Sync-C = how much it stands out from the other shifts (higher is better)

Sync-C ≈ 7–8 with Sync-D ≈ 6.5–7 is close to real video on lip-reading datasets (Wav2Lip, ACM MM 2020).

Reading the scoreboard (and its trap)

System	Benchmark	Sync-C ↑	Sync-D ↓
Real video	LRS2 (lip-reading set)	7.84	6.74
Wav2Lip (2020)	LRS2	7.79	6.39 ← beats real video?
NAVA (Baidu, 2026)	Verse-Bench	7.79	7.57
Ovi 1.1 (Character.AI)	Verse-Bench	7.48	7.98
LTX 2.3 (Lightricks)	Verse-Bench	7.25	7.69
daVinci-MagiHuman	Verse-Bench	7.15	7.82

Only compare numbers on the same benchmark. Wav2Lip was trained against a SyncNet-style judge, so it can "beat" reality on that judge. Sources: Wav2Lip arXiv 2008.10010; NAVA arXiv 2605.30073.

Game: which one is in sync?



Clip A

Watch his mouth on the hard consonants.

Clip A (re-timed from Veo 3) · click to watch · source ↗



Clip B

Same clip. Is anything different?

Clip B (re-timed from Veo 3) · click to watch · source ↗



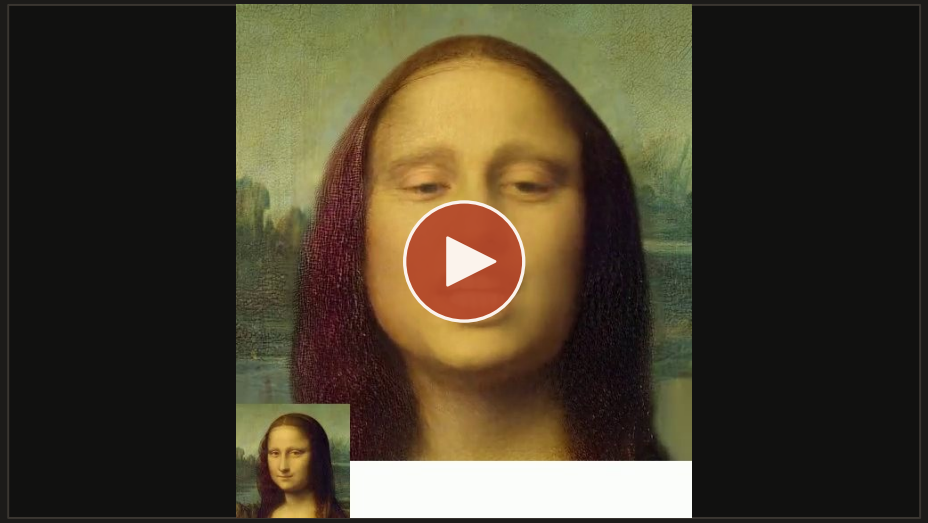
Clip C

And this one?

Clip C (re-timed from Veo 3) · click to watch · source ↗

Vote: A, B or C? Then: is any of them early rather than late?

VASA-1: a painting that talks



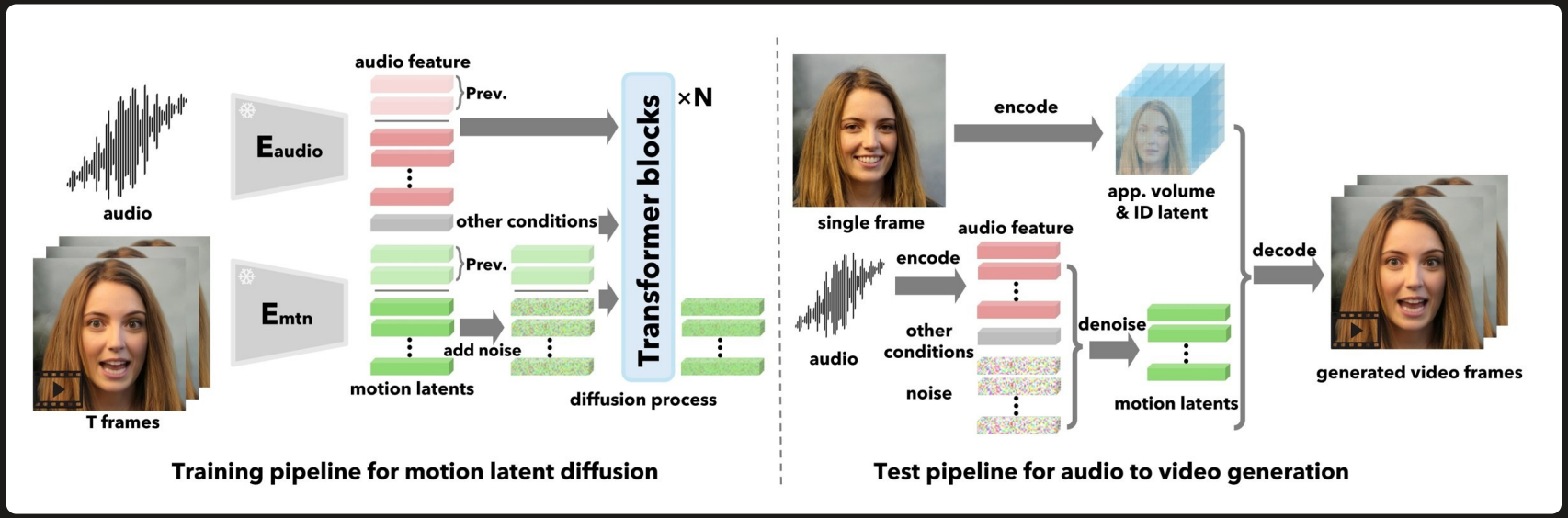
VASA-1 · Microsoft Research · click to watch

- One photo + one audio track → a talking, moving face
- Face latent separates identity from motion; a small diffusion transformer (~29M parameters) writes the motion
- 512×512 at up to 40 fps, ~170 ms latency on one RTX 4090
- Microsoft did not release it, citing misuse risks

▶ VASA-1 project page

vasavatar.github.io/VASA-1

How VASA-1 works: generate motion, not pixels



Left: training a diffusion transformer on motion latents, conditioned on audio and other signals. Right: one photo is encoded once into appearance and identity; only the motion is generated, then decoded into frames. Figure: VASA-1, Microsoft Research, arXiv 2404.10667.

Same face, same words, four emotions



VASA-1 lets you dial gaze, head distance and emotion as separate controls · VASA-1 · Microsoft Research · click to watch · source ↗

OmniHuman: from face to whole body



OmniHuman-1 · ByteDance · click to watch

- One image + audio → full-body performance: hands, posture, instrument
- Trained on 18.7K hours by mixing weak (text) and strong (audio, pose) conditions
- OmniHuman-1.5: a multimodal LLM writes an acting plan first, then the video model follows it
- ByteDance; available as a product, weights not released

▶ **OmniHuman-1 project page**

omnihuman-lab.github.io

▶ **OmniHuman-1.5 (arXiv 2508.19209)**

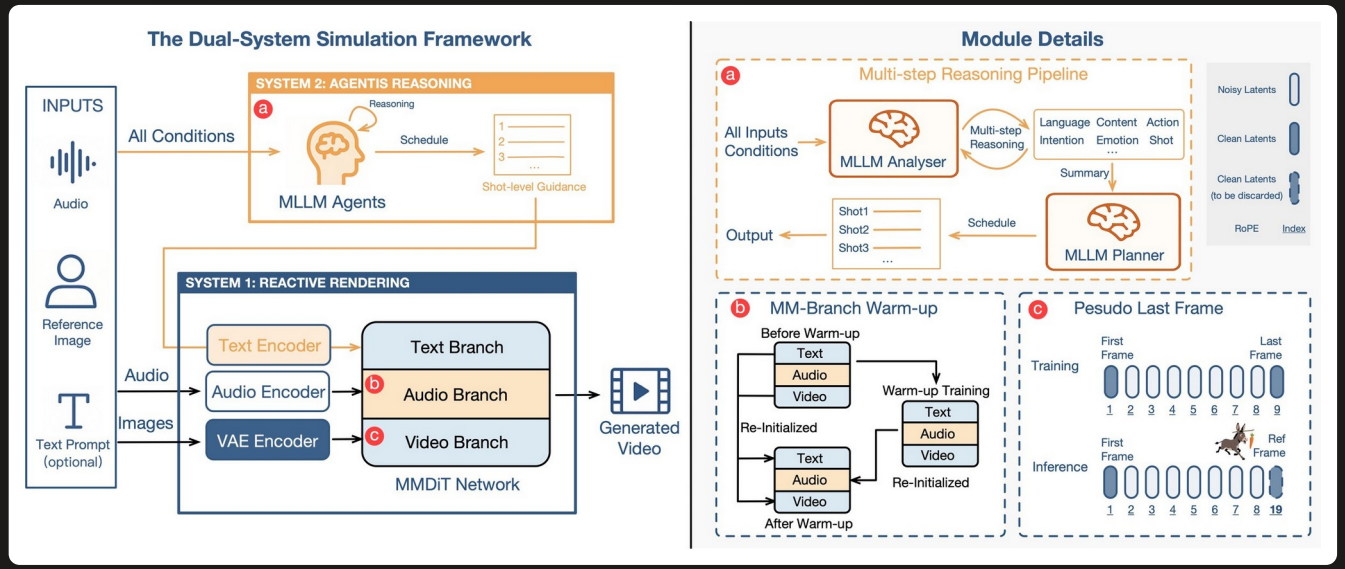
arxiv.org/abs/2508.19209

OmniHuman-1.5: acting, not just talking



Watch the eyes and the pauses, not only the lips · OmniHuman-1.5 · ByteDance · click to watch · source ↗

OmniHuman-1.5: a planner thinks, a renderer acts



"System 2": multimodal LLM agents analyse the audio, image and text and write a shot-by-shot plan. "System 1": an MMDiT with text, audio and video branches renders it. Figure: OmniHuman-1.5, ByteDance, arXiv 2508.19209.

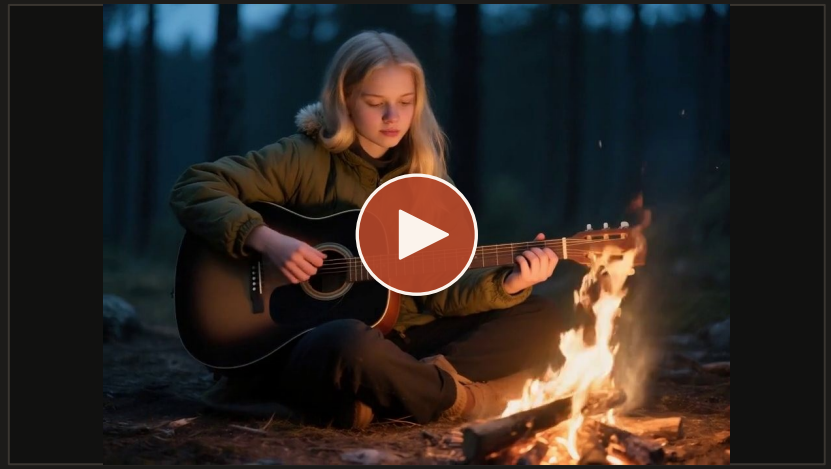
Singing: voice drives breath, body and instrument



Wan2.2-S2V (Alibaba)

Speech-to-video for film: a singer on a ship in a storm, camera and body moving with the song.

Wan2.2-S2V · Alibaba Tongyi · click to watch · source ↗

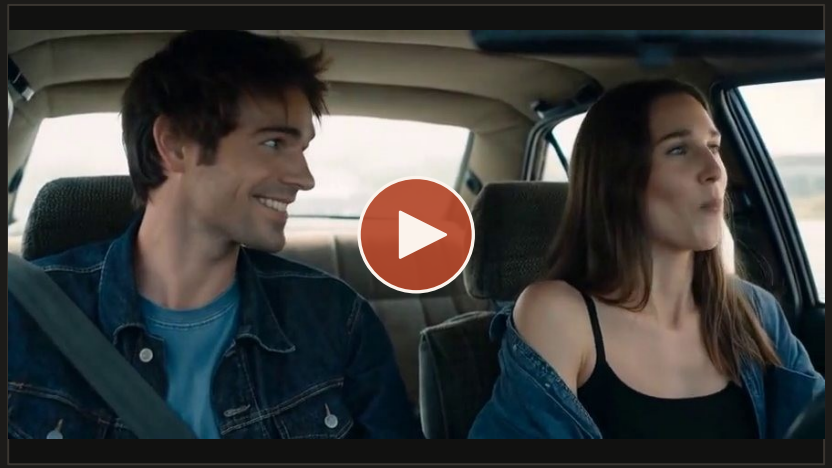


HunyuanVideo-Avatar (Tencent)

Strumming hands, breathing and firelight from one photo and a song.

HunyuanVideo-Avatar · Tencent · click to watch · source ↗

Two people: who is speaking?



MultiTalk (NeurIPS 2025)

Each voice track is bound to one person, so the right mouth moves.

MultiTalk · MeiGen-AI · click to watch · source ↗

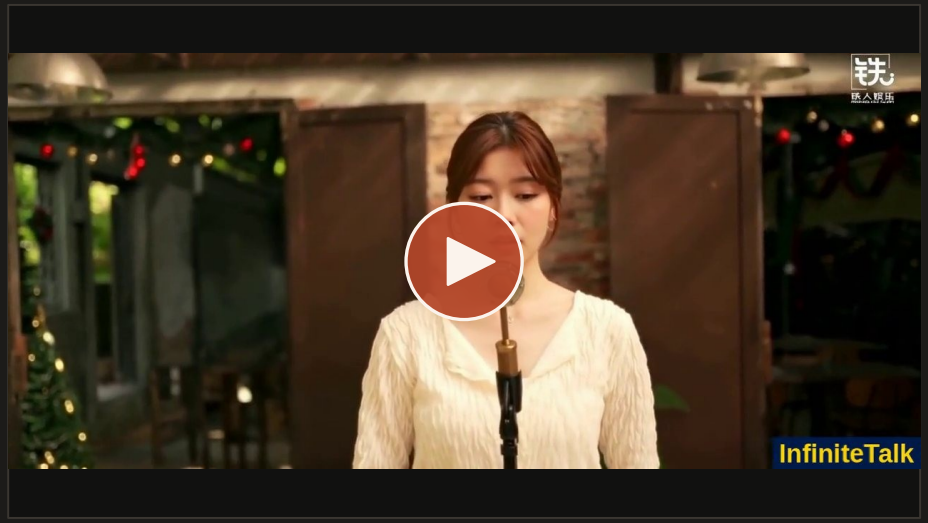


HunyuanVideo-Avatar (Tencent)

Multiple characters, with an emotion module for each voice.

HunyuanVideo-Avatar · Tencent · click to watch · source ↗

Dubbing: one performance, many languages



InfiniteTalk · MeiGen-AI · click to watch

- InfiniteTalk “sparse-frame dubbing”: re-generates lips, head, body and expression to a new voice track
- Unlimited length; open weights (Apache-2.0)
- Film precedent: Fall (2022) changed 30+ swear words with Flawless “vubbing” to reach PG-13
- Question: whose performance is the dubbed version?

▶ InfiniteTalk project page

meigen-ai.github.io/InfiniteTalk

▶ Flawless TrueSync on Fall (YouTube)

www.youtube.com/watch?v=iQ1OPpj8gPA

A performance built from references



Character, wardrobe, set and voice can each be pinned by a reference · Vidu · ShengShu (version not stated) · click to watch · source ↗

Directing with references

Who: identity references

Vidu Q4 takes up to 15 reference images to pin down a character, wardrobe and set (Oct 2026).

How they sound: voice references

Vidu Q4 and Seedance 2.0 accept reference audio so a voice stays the same across shots.

Bring your own track

Audio-to-video (LTX-2, Wan-S2V, InfiniteTalk): record the line yourself, let the model act it.

Which line is whose

NAVA ties a speaker embedding to each span of speech; MultiTalk binds one track per person.

Rule of thumb: the more you specify by reference, the less you leave to chance, and the more consent you need.

Directing the delivery: what to ask for

Direction	Voice	Face and body	Prompt words that help
Tired	Slower, lower, breathy	Heavy lids, slow blinks, little head motion	“quietly, exhausted, after a long pause”
Angry, held in	Clipped, even pitch	Tight jaw, fixed stare	“through clenched teeth, without raising his voice”
Nervous	Fast, rising ends, fillers	Gaze darts away, small swallows	“hesitant, um, avoids eye contact”
Delighted	Wider pitch range, laugh in voice	Raised cheeks, eyes crinkle	“can’t hide a smile, laughs on the last word”

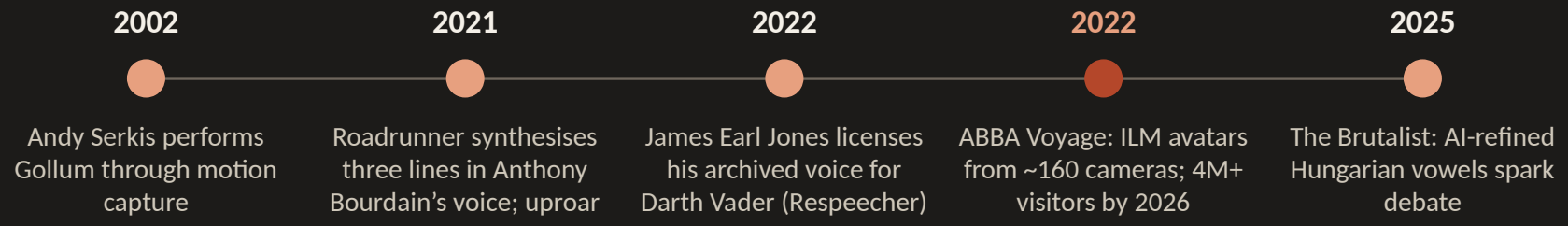
Actors call these “playable actions”. Describe what the character does, not only the emotion label.

EXERCISE · 5 MINUTES

One line, three deliveries: “I never said you could leave.”

In pairs, write three prompts that make this line a threat, a joke, and a plea. Describe the voice, the face and one gesture for each.

Performers have been digitised before



Break: student presentations

Presentations

This week’s sign-ups share ongoing or past work, about 10 minutes each.

10-minute break

Then Story 3: characters that listen in real time.

Try during the break

Play one demo from tonight on your phone with the sound off, then on. What changed?

STORY 3 OF 4

03

Characters that listen

From rendering a clip to holding a conversation, in real time



Two numbers that define “real time”

25 fps

Throughput: make video at least as fast as it plays. 40 ms per frame

~200 ms

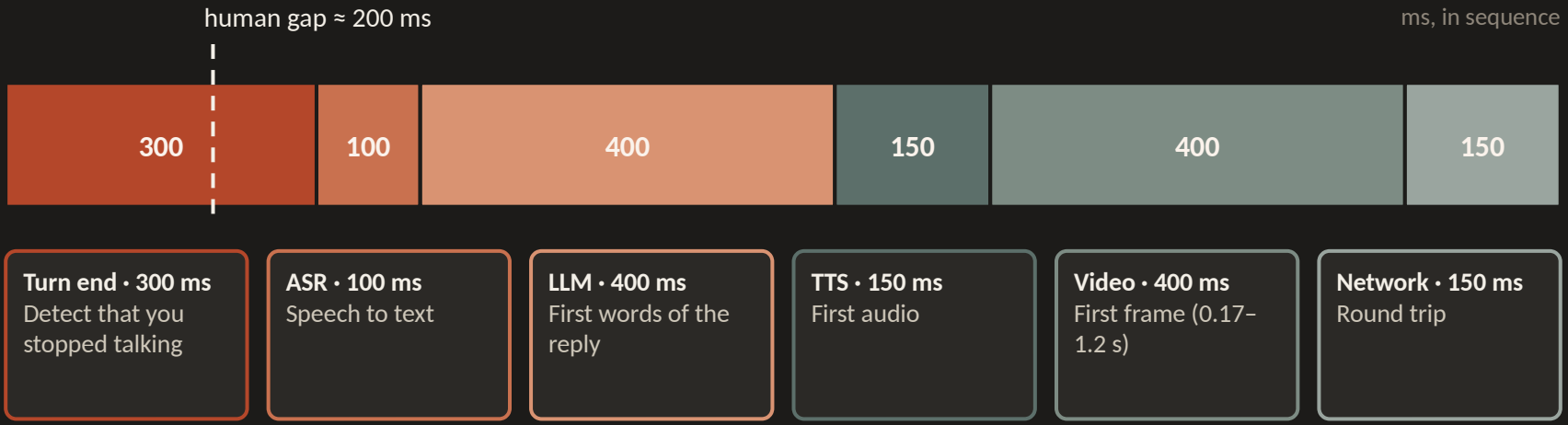
Latency: the typical gap between turns in human conversation

600 ms+

What it takes a human to plan a spoken reply, so we predict the end of your sentence

A model can be fast (high fps) and still feel slow (high latency). Conversation needs both. Source: Levinson & Torreira, Frontiers in Psychology 2015; Stivers et al., PNAS 2009 (10 languages).

The latency budget of a talking avatar



Run in sequence, a typical pipeline takes about 1.5 s, far from 200 ms. Fixes: overlap the stages (stream every step), predict the end of the turn, and make the video model answer in chunks. Voice-agent figures: Soniox latency guide; video range from the papers on the next slides.

Why a normal video diffusion model can’t talk back

50 steps

A typical sampler denoises 30–50 times

→

× 2

Classifier-free guidance runs each step twice (Week 3)

→

× all frames

Every frame attends to every other, past and future, so nothing is ready until the whole clip is

→

Seconds to minutes

Fine for a film; useless for a conversation

Three tricks you already know

Chunks + KV cache (Week 7)

Generate a few frames at a time, looking only backwards; cache the past so it is not recomputed. Self Forcing trains on the model's own frames to stop drift.

Fewer steps (distillation)

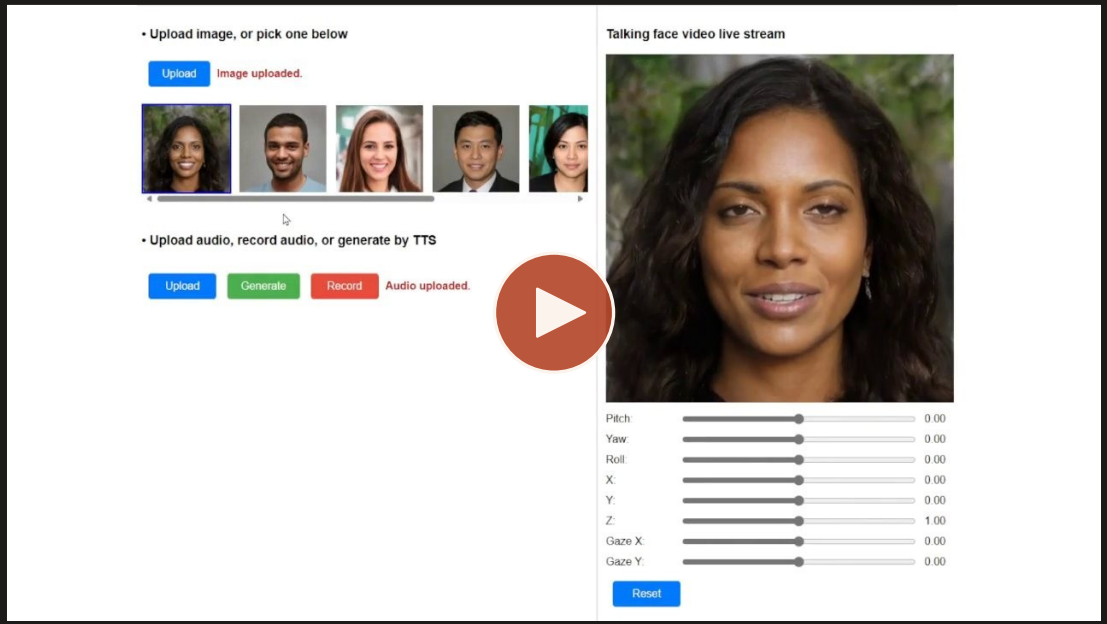
Train a student to jump to the answer in 1–4 steps. TalkingMachines: DMD distillation cuts 24 model evaluations to 2.

Fewer bits (Week 6)

Quantize weights and attention to 8 or 4 bits, as Yaofu showed, so the same GPU does more per millisecond.

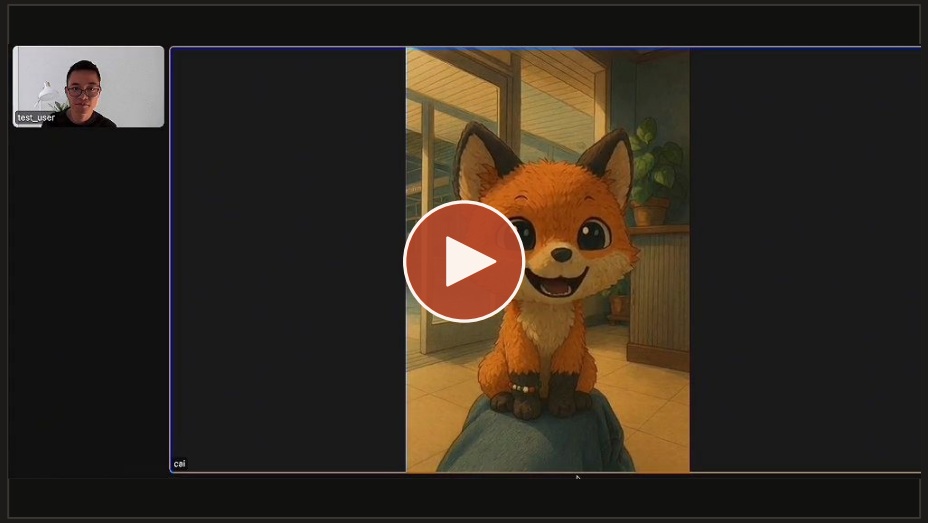
Plus a systems trick: put the transformer and the decoder (VAE) on different GPUs so they work in parallel.

VASA-1 streaming: a face in a live call (2024)



Up to 40 fps at 512×512 with ~170 ms delay, on one RTX 4090 · VASA-1 · Microsoft Research · click to watch · source ↗

TalkingMachines: a FaceTime call with a character



TalkingMachines · Character.AI · click to watch

- Character.AI turned Wan 2.1 (14B, image-to-video) into an 18B real-time avatar model
- Distillation: 24 model evaluations → 2 per chunk (DMD)
- Chunks of 3 latent frames; the KV cache keeps the first chunk (identity) and the previous one
- Any style: photo, cartoon, animal; paired with an audio LLM

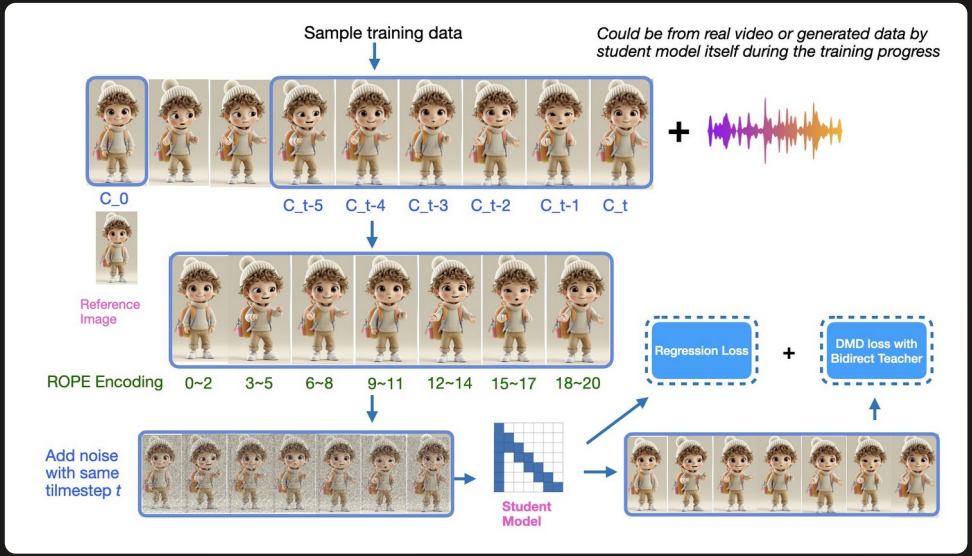
▶ **TalkingMachines project page**

aaxwaz.github.io/TalkingMachines

▶ **Paper (arXiv 2506.03099)**

arxiv.org/abs/2506.03099

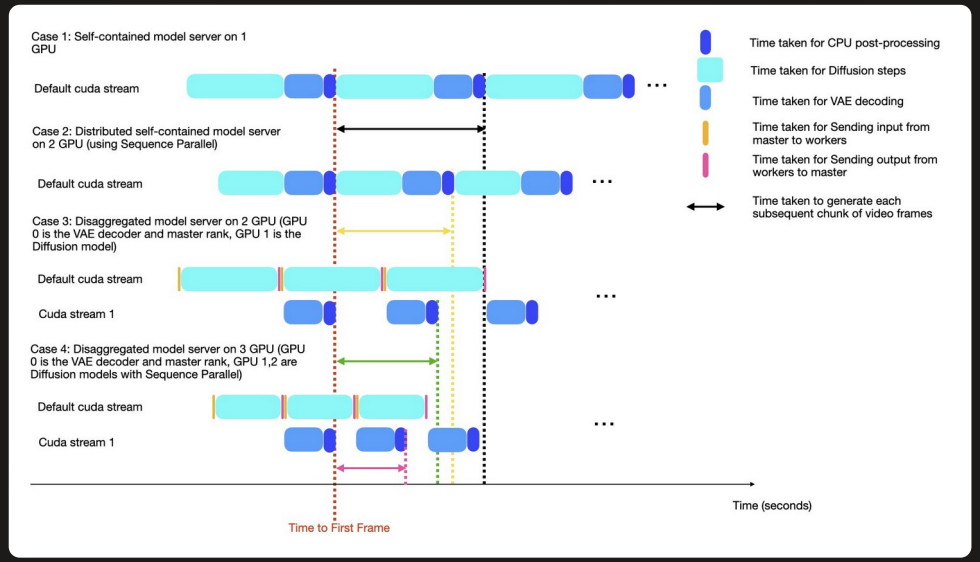
How streaming distillation works (TalkingMachines)



- Top: training clips are cut into chunks of 3 latent frames, plus the reference image (C_0)
- Each chunk only sees itself, the first chunk and the previous chunk: a sparse causal mask
- A slow bidirectional teacher (24 steps) supervises a fast causal student (2 steps)
- Training can even use the student's own generated video, like Self Forcing (Week 7)

Figure: TalkingMachines, Character.AI, arXiv 2506.03099 (2025).

Where the milliseconds go: split the work across GPUs



- Case 1: one GPU does diffusion, then VAE decoding, then waits
- Case 3: GPU 0 decodes (VAE) while GPU 1 already denoises the next chunk
- Case 4: more GPUs share the diffusion steps (sequence parallel)
- The red line, time to first frame, moves earlier with each design

Figure: TalkingMachines runtime analysis, Character.AI, arXiv 2506.03099.

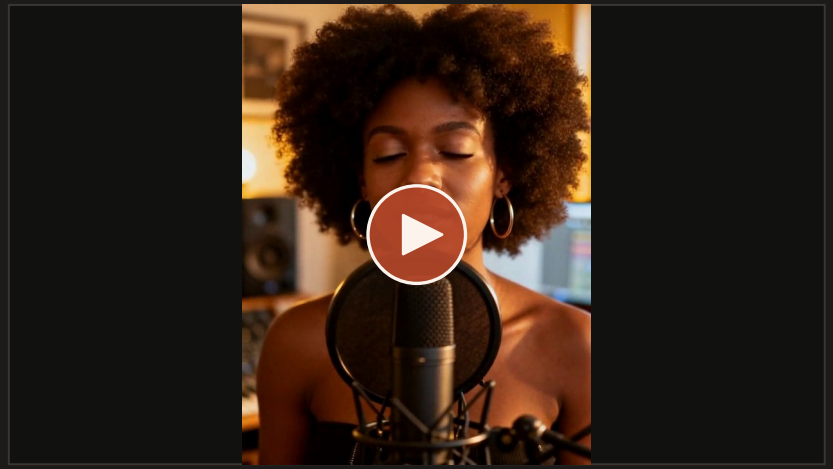
StreamAvatar: the whole upper body, live



Talking

Distilled from Wan2.2 into a 3-step streaming model: 25 fps, ~1.2 s delay on two H800s.

StreamAvatar · Tencent Hunyuan et al. · click to watch · source ↗

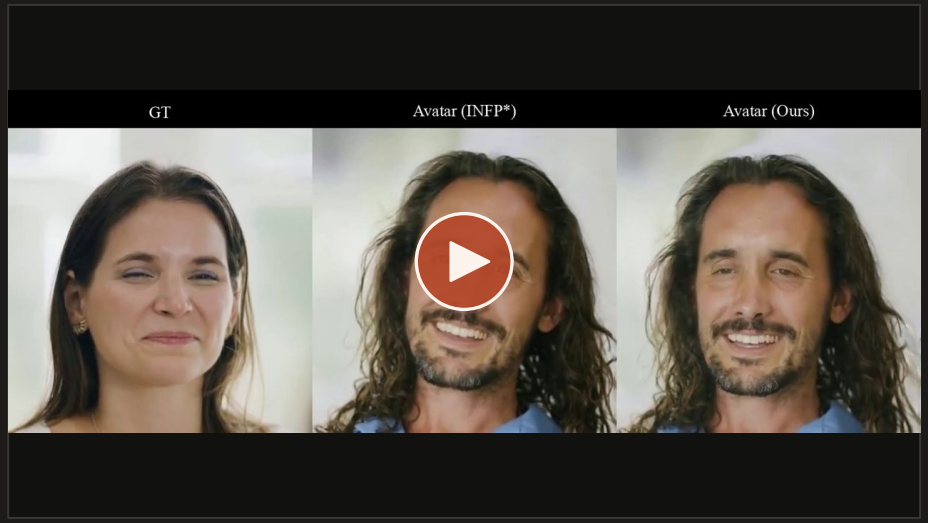


Singing

A “reference sink” keeps the first frames in memory so identity does not drift.

StreamAvatar · Tencent Hunyuan et al. · click to watch · source ↗

Avatar Forcing: an avatar that listens (CVPR 2026)



Avatar Forcing · KAIST, NTU · click to watch

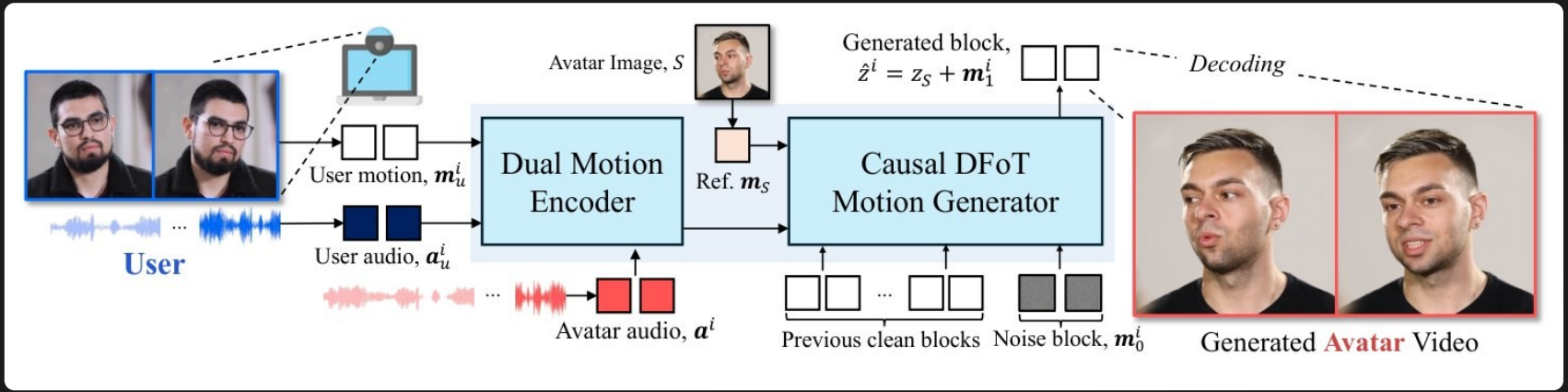
Columns: real person · previous method · Avatar Forcing

- Reacts while you talk: nods, smiles, laughs, using your audio and your motion as input
- Diffusion forcing: 10-frame blocks, 2 frames of look-ahead, rolling KV cache
- Learns good reactions without labels: the losing example is the same model with your input removed
- ~500 ms on one H100: 6.8× faster than INFP; preferred in over 80% of comparisons

▶ Project page (more listening demos)

taekyungki.github.io/AvatarForcing

Inside Avatar Forcing



The user's motion and audio and the avatar's own audio are encoded together (Dual Motion Encoder); a causal diffusion-forcing generator writes the next block of motion, conditioned on past clean blocks and a reference frame; a decoder turns motion into video. Figure: Avatar Forcing, arXiv 2601.00664 (CVPR 2026).

Real-time scoreboard, 2024–2026

System	Delay to respond	Speed	Hardware	Open?
VASA-1 (Microsoft, 2024)	~170 ms	40 fps	1× RTX 4090	No
Livatar-1 (Hedra, 2025)	0.17 s	141 fps	1× A10	No
Avatar Forcing (CVPR 2026)	~0.5 s	25 fps	1× H100	Page only
OmniForcing (2026), with sound	~0.7 s	~25 fps	1 GPU	Paper
Hallo-Live (2026), with sound	0.94 s	20 fps	2× H200	Yes
StreamAvatar (2025), body	~1.2 s	25 fps	2× H800	Not stated
Wan-Streamer v0.1 (2026)	~0.2 s + network	25 fps (192p)	2 GPUs	Not stated

Not directly comparable: resolutions, face-only vs body, and what counts as “delay” differ by paper. Sources: each paper’s arXiv page (see reading guide).

Pipeline or one brain?

Cascade (most products today)

- Separate models: turn detection → ASR → LLM → TTS → avatar
- Easy to swap parts and pick the best of each
- Every hand-off adds delay and loses tone of voice
- Typical: around 1 s or more

End-to-end (Wan-Streamer, 2026)

- One transformer takes in text, audio, video and puts out audio + video
- Streams in 160 ms units; a “thinker” passes its memory to a “performer”
- Can hear how you said it, not only what you said
- ~200 ms model-side; still low resolution

Wan-Streamer: one model hears, thinks and answers



A live call: the person in the small window talks; the character answers in voice and video · Wan-Streamer · Alibaba Wan · click to watch · source ↗

From our lab: TalkVid

1,244 h

of talking-head video, filtered from 6,000+ hours of web video

7,729

speakers, ages 0 to 60+, upper body included

15

languages (867 h English, 249 h Chinese); 87.6% at 1080p or above

500

TalkVid-Bench clips, balanced by age, gender, ethnicity and language

TalkVid: A Large-Scale Diversified Dataset for Audio-Driven Talking Head Synthesis (CVPR 2026 Findings). CUHK-Shenzhen, SYSU and HKUST. arXiv 2508.13618 · github.com/FreedomIntelligence/TalkVid

How TalkVid was filtered: 6,000+ hours → 1,244

Collect

30,000+ videos at 1080p or higher

→

Cut

Shot detection; drop clips under 5 s or without speech

→

Look

Aesthetic score (DOVER ≥ 7) for sharp, well-lit video

→

Move

Motion tracking: drop jittery and frozen clips

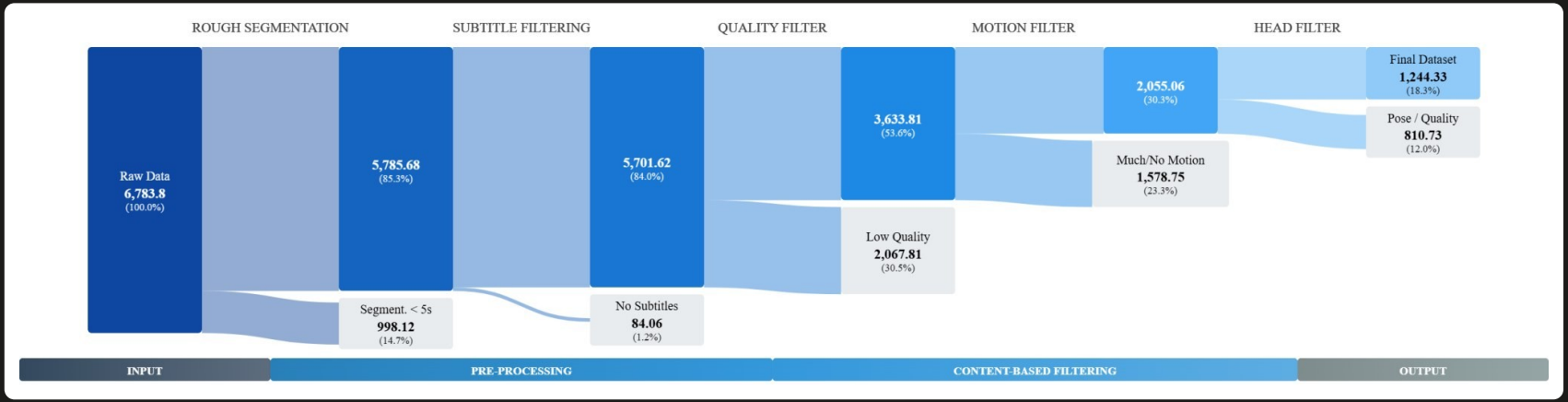
→

Check

Head pose and completeness; humans agree with the filters 95% of the time

Training on the balanced 160-hour TalkVid-Core gave lower FVD than training on older datasets (175.1 vs 188.7 HDTF / 183.2 Hallo3) on the HDTF test set.

The real filter funnel, and what it keeps



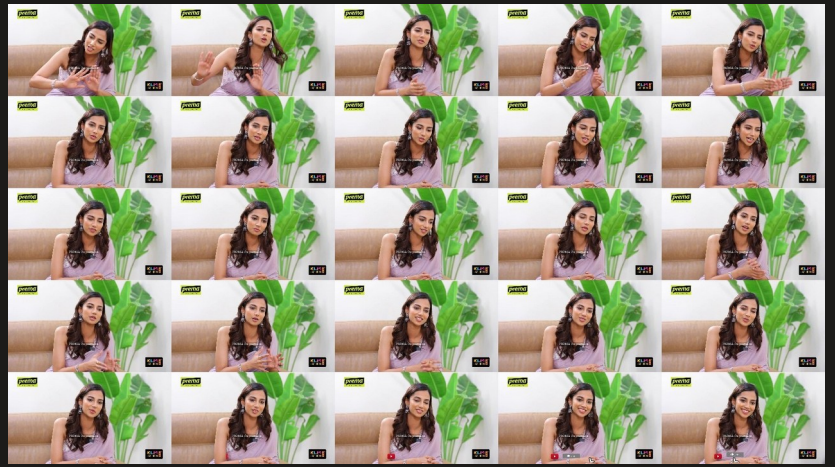
Raw hours on the left; segmentation, subtitle, quality, motion and head filters remove clips until 1,244 hours remain. Figure: TalkVid, arXiv 2508.13618 (CVPR 2026 Findings).

What the quality filter drops and keeps



Dropped: low aesthetic score

Dim, soft, low-contrast footage.



Kept: high aesthetic score

Sharp, well lit, stable framing.

Example frames from TalkVid's DOVER filter (arXiv 2508.13618). A model learns the look of its data, so this filter decides the look of every avatar trained on it.

What runs where (for your projects)

A laptop or mid GPU

MuseTalk: single-step mouth inpainting, 30+ fps on a V100-class GPU. LatentSync 1.5: about 8 GB of GPU memory (offline).

One RTX 4090 / 5090

Light real-time heads (e.g. SoulX-FlashHead Lite, reported 96 fps on a 4090). 14B avatar models at 480p offline, not live.

Several data-centre GPUs

Live 14B-class avatars with body and sound: StreamAvatar, Hallo-Live, FlashTalk (8× H800).

Real-time 14B-class avatars on one consumer GPU is an open problem, and one our lab works on: it needs distillation, chunked generation and low-bit kernels together.

Where live characters already are

Taxis in Japan

ESTsoft's "Perso Interactive" avatar signed a deal with NTT and taxi operator Nihon Kotsu (Feb 2026) to interpret and guide tourists.

Creator tools

ElevenLabs added avatars to its speech platform (June 2026): a script becomes a talking-head video in one place.

Companions and characters

Character.AI builds the research (Ovi, TalkingMachines) for video calls with fictional characters.

Museums and performance

ABBA Voyage shows digital performers can fill an arena; live, responsive versions are the next step.

Artists working with their own likeness

xhairymutantx · Whitney Biennial 2024

Holly Herndon and Mat Dryhurst trained a model on costumed images of Herndon, so any prompt produces a distorted “her”: a likeness the artist controls.

The Call · Serpentine, 2024–25

Herndon and Dryhurst recorded 15 UK choirs, built singing models with Ircam, and set up a data trust so the choristers share control.

A pattern for your own projects: use your own voice and face, or people who agreed, and make the consent part of the work.

FOR THE ROOM

Would you talk to a live avatar of your teacher? Of someone who has died?

What would make it feel respectful, and what would make it feel wrong?

STORY 4 OF 4

04

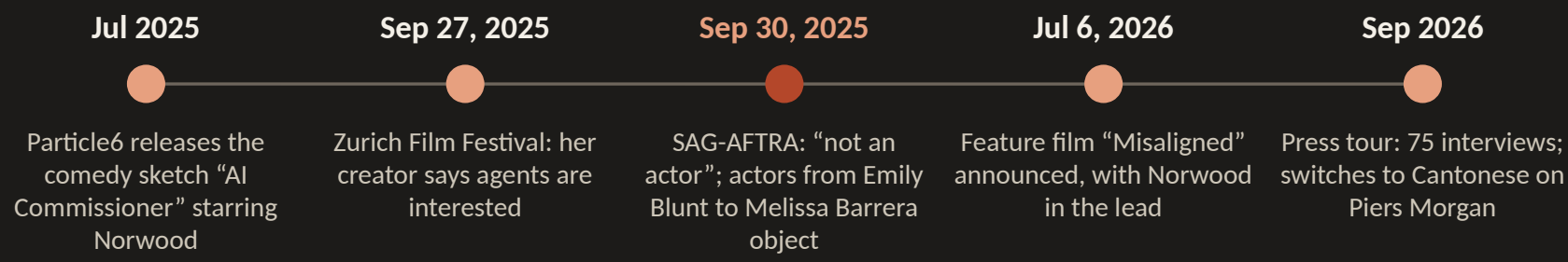
Who is the performer?

Synthetic actors, digital replicas, and the new rules of 2026



OmniHuman-1.5 · ByteDance

Tilly Norwood: an “actress” who never existed



Watch: synthetic performers, from sketch to stage

Tilly Norwood · “AI Commissioner”

The July 2025 comedy sketch that introduced her (Particle6).

▶ Play on YouTube

Tilly Norwood · “Take The Lead”

Her March 2026 music video answering the backlash; 18 human credits, AI vocals.

▶ Play on YouTube

ABBA Voyage

Digital performers with the band’s consent and a live band: 4M+ visitors.

▶ Official trailer

Holly Herndon · Holly+

The artist’s own voice model, offered to others on her terms.

▶ “Jolene (feat. Holly+)”

Opens in the browser. Same technology, four different answers to “who agreed?”

Two positions, in their own terms

SAG-AFTRA (Sep 30, 2025)

- Norwood is a character generated by a program, not an actor
- Trained on countless performers' work without permission or pay
- Has no life experience or emotion to draw on
- Producers must notify and bargain with the union before using a synthetic performer

Her creator (Particle6)

- A creative work, "a piece of art", not a replacement for a human
- AI is a new tool, like animation, puppetry or CGI
- Says she has no intention of replacing human actors
- Plans dozens more characters in a "Tillyverse"

What the actors won: the 2026 SAG-AFTRA contract

Synthetic performers

Not used in a human role unless they add “significant additional value”; producers must go to the union first.

Digital replicas

Informed consent and fair pay for replicas of living or deceased performers; a business reason is needed before scanning.

Strikes and minors

No replica may replace a performer during a strike; extra protections for minors.

Ratified June 4, 2026 (91.4% yes); runs July 2026 – June 2030. The union’s negotiator called it a strong disincentive, not a ban. Video-game actors won similar consent rules after an 11-month strike (2024–25).

Source: LA Magazine; TheWrap; Malay Mail/AFP (May–June 2026)

The law, region by region (October 2026)

Where	Rule	Status
California	AB 2602 voids vague digital-replica clauses; AB 1836 needs an estate's consent for replicas of the dead	In force since Jan 2025
United States	NO FAKES Act: a federal right to your voice and likeness, with takedowns	Not law; blocked Sep 30, 2026
Mainland China	AI-generated content must carry visible labels and hidden metadata labels	In force since Sep 1, 2025
European Union	AI Act Art. 50: deepfakes must be disclosed; outputs machine-marked	Applies from Aug 2, 2026
Denmark	Copyright-style protection of your own face and voice against deepfakes	Bill in parliament, Oct 2026

Not legal advice. For your work: label AI-generated people and voices, and get consent from anyone recognisable.

Your voice is part of your identity

Lehrman & Sage v. Lovo (2025)

Two voice actors were paid a few hundred dollars for “research” recordings that became commercial clone voices. A New York court let their publicity-rights and contract claims go ahead.

“Sky” (2024)

Scarlett Johansson declined to voice ChatGPT; a voice that sounded like her appeared anyway. OpenAI paused it, saying it was another actress’s natural voice.

Darth Vader in Fortnite (2025)

Even with James Earl Jones’s estate’s consent, the union filed a complaint that an AI voice replaced paid actors.

What audiences say (NRG × TheWrap, March 2026)

56%

say AI actors will never be as good as human actors

7%

say AI actors are already as good

~3 in 5

say releasing a film with an AI-written script is unacceptable

Yet the largest group said AI use would not change their interest in a film. Audiences judge the result, and the story about how it was made.

Two flashpoints from this year

Feb 2026 · The Seedance 2.0 clip

A viral fight scene with two famous actors' likenesses, made in Seedance 2.0, drew condemnation from the Motion Picture Association and SAG-AFTRA. ByteDance pledged fixes.

Jun 2026 · Dreams of Violets at Tribeca

Ash Koosha's film, made with Kling, Claude and Gemini about this year's protests in Tehran, was billed as the first fully AI-generated feature at a major festival.

Same tools, opposite receptions. The difference was not the model; it was whose likeness, whose story, and who agreed.

A checklist for your Assignments 2 and 3

Consent

Only clone or animate a real person’s face or voice if they agreed, in writing. Yourself is easiest.

Label

Say clearly which faces, voices and sounds are generated (and you must, in mainland China and the EU).

Credit

Name the models, the voice references and every human who performed, recorded or wrote.

Keep sources

Prompts, seeds, reference audio, model versions, raw takes (Week 7’s rule).

DEBATE · 10 MINUTES

“An AI performer that never existed should be allowed to win an acting award.”

Left side of the room argues for; right side against

Two speakers per side, one minute each

Then the whole room votes by show of hands

Arguments to start from

For

- Animated performances already move us; awards could judge the result, not the method
- Directors, writers and voice references made the choices; the award could go to them
- A separate category could honour new craft without displacing actors

Against

- Acting awards honour a person's lived skill and risk on screen
- The "performance" is averaged from thousands of uncredited actors
- Prizes for synthetics reward replacing the people the model learned from

Four stories, one line each

01

Sound

Picture and sound now share one model and one clock; the human job is choosing what to hear.

02

Performance

Direct with references and playable actions; check the p, b and m sounds.

03

Real time

Chunks, fewer steps and fewer bits turn a renderer into a conversation partner that listens.

04

Performer

Consent, credit and labels are now contract and law, not just good manners.

Tonight → your assignments (both due Nov 30)

Assignment 2 · Reverse, Rebuild, Remix

When you reverse-engineer an AI video, reverse-engineer its sound too: which layers were generated, and in which model? Then remix with a native-audio model or your own voice track.

Assignment 3 · Make Your Own Game

Give one character a voice and a face that reacts. A talking portrait (MuseTalk, LatentSync) or a short generated cut-scene with dialogue is enough.

Use your own voice, or a friend who agreed in writing. Label what is generated.

Three readings for this week

01

LTX-2: Efficient Joint Audio-Visual Foundation Model

Technique · Lightricks, Jan 2026. Abstract, architecture figure, two demos.

arXiv 2601.03233

02

The Call · Holly Herndon & Mat Dryhurst

Art + tech · Serpentine, 2024–25. Choirs, singing models, and a data trust.

serpentinegalleries.org

03

Tilly Norwood to lead her first film

Culture · Variety, July 2026, with the SAG-AFTRA statement.

variety.com

+

Optional: Avatar Forcing; TalkVid

Technique · real-time listening avatars, and the data behind them.

arXiv 2601.00664 · 2508.13618

Quiz 08 · five short questions

Two technique questions

Joint audio-video vs adding sound afterwards; the tricks that make avatars real-time.

Two art questions

Synchresis and why lip-sync errors are so visible; directing a line beyond the words.

One reading reflection

Any one of the three readings: its main idea, and whether you agree.

hyang.org/amcc5160/quiz-08 · 4 points each · due at the end of the semester, editable until then.

Every demo from tonight, in one place

Story	Demo	Where to find it
1 · Sound	Veo 3, Ovi, LTX-2, UniVerse-1	deepmind.google/models/veo · aaxwaz.github.io/Ovi · github.com/Lightricks/LTX-2
2 · Performance	VASA-1, OmniHuman, MultiTalk, HunyuanVideo-Avatar, InfiniteTalk	microsoft.com/…/vasa-1 · omnihuman-lab.github.io · github.com/MeiGen-AI
3 · Real time	VASA-1 live, TalkingMachines, Avatar Forcing	arxiv.org/abs/2506.03099 · taekyungki.github.io/AvatarForcing
4 · Performer	Tilly Norwood, SAG-AFTRA, The Call	variety.com · sagaftra.org · serpentinegalleries.org

All links are also on the Lecture 08 page, under “Demos”. Videos belong to their creators and are shown for teaching.

Next week: critique, evaluation, provenance

Metrics vs. eyes

Tonight's trap, Wav2Lip "beating" real video, generalised: what do scores really measure?

Studio critique

Bring one generated clip with sound; we critique it like a film crit.

Provenance

Model cards, watermarks, labels and disclosure: how to release responsibly.

Sound, performance, presence

Generated video learned to speak, to act, and to listen. Choose what we hear, direct how it is said, and ask whose voice it is.

Sources: Google DeepMind Veo 3 model card; LTX-2 (arXiv 2601.03233); Ovi (arXiv 2510.01284); Kandinsky 6.0 (arXiv 2610.05608); NAVA (arXiv 2605.30073); OmniForcing (arXiv 2603.11647); Wav2Lip (arXiv 2008.10010); VASA-1 (arXiv 2404.10667); OmniHuman (arXiv 2502.01061, 2508.19209); TalkingMachines (arXiv 2506.03099); Avatar Forcing (arXiv 2601.00664); Livatar-1 (arXiv 2507.18649); Wan-Streamer (arXiv 2606.25041); TalkVid (arXiv 2508.13618); CMU 10-423 Lecture 24; Levinson & Torreira 2015; SAG-AFTRA, Variety, TheWrap, LA Magazine, NRG. Demo videos belong to their creators, shown for teaching.

APPENDIX

A

Do it yourself

Tools, a workflow, prompts and checks for Assignments 2 and 3

Toolkit: open models for each job

Job	Open model	Size / hardware (as reported)	Licence
Picture + sound together	LTX-2 (Lightricks)	19B; one large GPU; ComfyUI support	LTX-2 Community
Picture + sound, lighter	Kandinsky 6.0 Lite · NAVA	3B · 6.3B	MIT · Apache-2.0
Two-person dialogue	Ovi · MultiTalk	11B · 480p on one RTX 4090 (low-VRAM mode)	Apache-2.0
Add Foley to a silent clip	MMAudio · HunyuanVideo-Foley	MMAudio from 157M: 8 s of audio in ~1.2 s	MMAudio weights non-commercial
Lip-sync an existing video	LatentSync 1.5 · MuseTalk	~8 GB GPU memory · 30+ fps (256×256 face)	Apache-2.0 · open
Re-dub a whole video	InfiniteTalk	480p/720p; unlimited length	Apache-2.0
Talking or singing body	Wan2.2-S2V · HunyuanVideo-Avatar	14B · 24 GB min (10 GB with TeaCache)	Apache-2.0 · Tencent licence

Check each licence before commercial use. Closed options with native sound: Veo 3.1, Seedance 2.0, Kling, Vidu, MiniMax (via their apps or APIs).

Workflow: a talking character in one evening

1 · Voice

Record the line yourself on your phone, in a quiet room. Timing is your performance

→

2 · Face

Generate or paint one image of the character, front-facing, mouth visible, good light

→

3 · Animate

Audio-to-video (LTX-2, Wan2.2-S2V, InfiniteTalk) or lip-sync (LatentSync, MuseTalk)

→

4 · Room

Add ambience and Foley (MMAudio) under the voice; keep the voice on top

→

5 · Check

Run the review checklist; fix and re-generate only the failing shot

Your own voice avoids every consent problem in Story 4, and gives you exact control of timing.

Review checklist for a generated performance

Lips

Do the lips fully close on p, b, m? Do teeth touch the lip on f, v?

Timing

Scrub frame by frame at a hard consonant or a hit: does the sound land on that frame?

Voice

Same voice across shots? Breaths and pauses where a person would take them?

Eyes and body

Is the gaze motivated? Do hands play the right rhythm on instruments?

Room

Does the reverb match the space? Does ambience continue across cuts?

Credits and labels

Models, voice references, consent and the AI label written down?

Measure sync yourself (two commands)

Score lip-sync with the reference SyncNet (github.com/joonson/syncnet_python)

```
sh download_model.sh
python run_pipeline.py --videofile my_clip.mp4 --reference take1 --data_dir out/
python run_syncnet.py  --videofile my_clip.mp4 --reference take1 --data_dir out/
# prints AV offset (frames), min distance (Sync-D) and confidence (Sync-C)
```

Feel a delay: make a copy with the sound 125 ms late (ffmpeg)

```
ffmpeg -i my_clip.mp4 -af "adelay=125:all=1" -c:v copy late_125ms.mp4
# compare the two side by side; try 45 ms early and 125 ms late
```

If the reported offset is not 0, shift your audio by that many frames (1 frame = 40 ms at 25 fps) and re-check.